

RESEARCH INTERESTS

Human–Computer Interaction; Computational Design; Digital Fabrication; Shape-Changing & Pneumatic Interfaces; Interactive Materials; Soft Sensing

EDUCATION

National University of Singapore, Singapore **Aug 2025–Jan 2027 (Expected)**
Master of Science in Engineering Design & Innovation (GPA: 4.39/5.00)

Zhejiang University, China **Sep 2021–Jun 2025**
Bachelor of Agricultural Sciences in Landscape Gardening (GPA: 3.90/4.00)
• **Honors & Awards:** ZJU Third Class Scholarship, Academic Excellence Honor (two consecutive years)

PUBLICATIONS & MANUSCRIPTS

Super-Resolution Tactile Sensing through Geometric Encoding in Liquid-Metal Microchannels

Shaoyu Cai†, Jiarui Feng†, **Tingxiao Wu†**, Hao Zheng, Xu Cheng, Kening Zhu, and Benjamin C. K. Tee
Major revision requested at npj Flexible Electronics

† Equal contribution; co-first authors

PneuStep: A Soft Pneumatic-Based Foot Haptic Device for Walking Experience on Different Terrains in VR

Huzhaorui Kang, Zhuolun Li, **Tingxiao Wu**, et al. *SIGGRAPH Asia 2026 Emerging Technologies, accepted.*

Breatho: Real-Time Respiratory Guidance System for Amateur Runners

Yuheng Tao, **Tingxiao Wu**, Shaoyu Cai. *SIGGRAPH Asia 2026 Emerging Technologies, accepted.*

RESEARCH EXPERIENCE

Immersive Reality Lab, National University of Singapore

Student Research Assistant | Aug 2025–Present | Supervised by Dr Shaoyu Cai

- **VacGuide (leading; ongoing):** Investigating how passive internal guide geometry programs angular motion through vacuum-driven membrane collapse; developed 5-mm open-guide and multi-guide primitives spanning 0–180° rotations within a shared pneumatic pouch, and is characterizing geometry–motion relationships toward inverse design of articulated shape-changing structures.
- **Capacitive Super-Resolution Sensing (co-leading; ongoing):** Developing a three-channel capacitive sensor for super-resolution contact localization on 3D surfaces; using greedy topology search to maximize three-channel response separability while reducing spatial aliasing, followed by machine-learning-based localization.
- **Liquid-Metal Super-Resolution Tactile Sensing:** Designed a family of geometrically encoded four-channel liquid-metal microchannel topologies for super-resolution touch and force sensing; established the sensor fabrication workflow and achieved a 56.6× SR factor, with 0.235 mm localization RMSE and 0.044 N force RMSE; quantified sensitivity and hysteresis and implemented interactive demonstrations.

NEXT Lab, Zhejiang University

Research Intern | May 2024–May 2025 | Supervised by Prof. Kejun Zhang

- Contributed to **ArchiTone**, a LEGO-inspired interactive system for visualized music education, including system prototyping and interaction design; resulting work available as an arXiv preprint (arXiv:2410.15273, 2024).

WORK EXPERIENCE

Zhejiang Hikrobotics Intelligent Tech – Experience Designer (UX Research & Prototyping) | **Jul–Nov 2024**
Shanghai Able Digital & Tech (Zhihuishu) – Design Intern | **Jun–Jul 2024**

SKILLS

- Prototyping & Fabrication: Pneumatic prototyping, 3D printing, heat sealing, soldering, electronics integration, rapid prototyping
- Hardware & Embedded: Arduino/ESP32, sensors, circuit prototyping
- Programming & Design: Python, C/C++, MATLAB, SolidWorks, Rhino, Unity, Figma